

Kaixiang Yao

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EDUCATION

Southern University of Science and Technology

Aug. 2023 - Jun. 2027

B.Eng. in Robotic Engineering

Advisor: Prof. Chenglong Fu

University of California, Berkeley

Jun. 2026 - Aug. 2026

Visiting Undergraduate Student

Advisor: Prof. Masayoshi Tomizuka

PUBLICATIONS AND MANUSCRIPTS

[1] Quasi-real-time digital twin for robotic assembly: a physics-based multi-fidelity framework (*Journal of Intelligent Manufacturing*, 2026)

Kaixiang Yao, Hao Chi, Dingye Chen, Jiwen Zhang

[2] TOGEARI: Interaction-Space Preconditioning for Condensed Finite-Element Systems with IPC Contact (*arXiv:2608.14162*, 2026)

Yanlin Liu, Chao Huang, Kaixiang Yao, Yao Shen

[3] Gaze-Driven Perception-Action Control of Multi-Channel FES for Efficient Grasp Restoration (*under revision, IEEE T-MRB*, 2026)

Kaixiang Yao, Zeyang Yang, Jinxin Sun, Kong Hoi Cheng, Junyun Fu, Zhaokai Chen, Chengyu Lin, Chenglong Fu

RESEARCH EXPERIENCE

University of California, Berkeley

Jun. 2026 - Present

Visiting Undergraduate Researcher

Advisor: Prof. Masayoshi Tomizuka; Mentor: Yuxin Chen

Embodied AI for Physical Manipulation

- Developing simulation-based evaluation workflows for learning and control in contact-rich manipulation.
- Studying how physical interaction feedback can support robust behavior under changing dynamics.
- Designing controlled evaluations to identify failure modes and improve experimental rigor.

Shanghai Jiao Tong University

Oct. 2025 - Present

Undergraduate Researcher

Advisor: Prof. Yao Shen

Robot Learning for High-Fidelity Rigid-Soft Contact

- Developed a simulation-and-learning framework for a four-finger PneuNet soft gripper and fragile-object grasping.
- Designed soft-gripper assemblies, pneumatic actuation semantics, contact/tactile definitions, and RL-compatible tasks.
- Studied how contact fidelity and force-safety rewards affect damage-free grasping.

Southern University of Science and Technology

Jul. 2024 - May 2026

Undergraduate Researcher

Advisor: Prof. Chenglong Fu

Eye-Tracking and Electrical Stimulation-Assisted Grasping

- Proposed normalized gaze-variance features and random-forest recognition with dual-threshold temporal filtering.
- Integrated YOLOv5 object detection and MediaPipe hand-keypoint estimation for object-aware grasping strategies.
- Designed an 8-channel, 54-electrode high-density FES sleeve for selective muscle activation.
- Verified approximately constant interaction time as gesture options increased, supporting O(1)-style selection behavior.

RESEARCH EXPERIENCE (CONTINUED)

Tsinghua University	Jun. 2025 - Aug. 2025
Undergraduate Researcher	Advisor: Prof. Jiwen Zhang
<i>Digital Twin for Robotic Assembly</i>	
• Built a ROS 2 non-blocking pipeline separating real-time safety control from background FEA. • Used kinematic TIE constraints and mesh-scale analysis to improve robustness and computational predictability. • Integrated full-field stress outputs into an FSM-based safety evaluator.	

TECHNICAL SKILLS

Programming: Python, Java, MATLAB, ROS 2
Robotics and Control: Robot modeling and control, grasp planning, safety reflex control, FSM decision logic
Learning and Perception: Reinforcement learning, YOLOv5, MediaPipe, random forests
Simulation and Modeling: MuJoCo/MJLab, FEA/FEM, rigid-soft contact, mesh-scale analysis
Design and Fabrication: CAD assemblies, PneuNet grippers, high-density FES sleeves

HONORS AND AWARDS

First Prize (Provincial Level, 2nd Place)	2025
Guangdong Provincial College Students Engineering Practice and Innovation Competition	
Outstanding Student Scholarship	2024 - 2025
Outstanding Student Scholarship	2023 - 2024

RESEARCH INTERESTS

My research interests are in robot learning, embodied AI, and physical manipulation, with an emphasis on reliable interaction with the physical world. I am particularly interested in combining simulation, mechanics, perception, and control to build adaptive robotic systems.